End Semester Examination 2025

M.TECH *Semester: First*

Name of the Course: **(Electric Vehicle Technology)** Course Code: MEV-101

Time: 3 Hour Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer **any two of the sub questions** among a,b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(10 marks)**	**CO1**
(a)	Use Gauss–Seidel method to solve the following system of linear equations: $28x+4y-z=32$, $2x+17y+4z=35$, $x+3y+10z=24$	
(b)	Using Newton Raphson Method, Find the root of the equation x sinx + cosx = 4.77 up to three decimal places.	
(c)	Solve $x^4-5x^3+20x^2-40x+60=0$, given that all the roots of f(x) = 0 are complex, by using Lin Bairstow method.	
Q2	(10 marks)	**CO2**
(a)	Using Relaxation method solve the following system of linear equations: $-9x+3y+4z=-100$, $x-7y+3z=-80$, $2x+3y-5z==-60$	
(b)	Define Cubic spline. Find the cubic spline corresponding to the interval (2,3) from the following table.: Hence compute (i) y (2.5) and (ii) y'(3).	

x:	1	2	3	4	5
y:	30	15	32	18	25

(c)	Solve the system of equation using LU decomposition method $x+y-z=2$, $2x+3y+5z=-3$, $3x+2y-3z=6$	
Q3		**CO3**
(a)	Using Jacobi's method, find all the eigen values and eigen vectors of the matrix, by Jacobi method: $\begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$	

(b)	Transform the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ to the tri-diagonal form using Given's method. Hence find the largest eigen value and the corresponding eigen vector of the tri-diagonal matrix.	
(c)	Normalized the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$ by singular value decomposition method.	
Q4	**(10 marks)**	
(a)	Solve the following LPP by graphical method. Minimize $Z = 20x_1 + 10x_2$ Subject to, $x_1 + 2x_2 \leq 40$ $3x_1 + x_2 \geq 30$ $4x_1 + 3x_2 \geq 30$ $x_1, x_2 \geq 0$	
(b)	Using Simplex method solve the following LP problem Minimize $Z = x_1 - 3x_2 + 2x_3$ Subject to, $3x_1 - x_2 + 2x_3 \leq 7$ $-2x_1 + 4x_2 \leq 12$ $-4x_1 + 3x_2 + 8x_3 \leq 10$ $x_1, x_2, x_3 \geq 0$	**CO4**
(c)	Using Big M-method solve the following LP problem Maximize $Z = 6x_1 + 4x_2$ Subject to, $2x_1 + 3x_2 \leq 30$ $3x_1 + 2x_2 \leq 24$ $x_1 + x_2 \geq 3$ $x_1, x_2 \geq 0$	
Q5		
(a)	Which of the following subsets S of R^4 are linearly dependent? (i) S = {(1, 0, 0, 0), (1, 1, 0, 0), (1, 1, 1, 1), (0, 0, 1, 1)}. (ii) S = {(1, –1, 2, 0), (1, 1, 2, 0), (3, 0, 0, 1), (2, 1, –1, 0)}. (iii) S = {(1, 1, 2, 4), (2, –1, –5, 2), (1, –1, 4, 0), (2, 1, 1, 6)}.	
(b)	Determine the range and null space of the linear transformation $T : R^3 \rightarrow R^4$ with $T(x, y, z) = (x - y + z, y - z, x, 2x - 5y + 5z)$.	**CO5**
(c)	Let $T: R^2 \rightarrow R^2$ be a linear transformation with T ((3, 4)) = (0, 1), T ((–1, 1)) = (2, 3). Find the matrix representation T [B, B] of T with respect to the ordered basis B = ((1, 0), (1, 1)) of R^2.	